🎓 HACKATHON PRESENTATION 2026

# SchemeMatch

AI-Powered, Conflict-Aware Government Scholarship Optimization Platform for Indian Students

🤖 AI Profile Extraction | ⚡ <50ms Evaluation | 🧱 MWIS Algorithm | 🌐 Offline-First

Tamil Nadu Focus · 30+ Schemes · Vanilla JS + Node.js + MongoDB

THE PROBLEM

# Why Deserving Students Lose Financial Aid

**₹1,200 Cr** in state and central scholarships go **unclaimed or rejected every year** — not because students don't want them, but because navigating government portals feels like preparing for the civil services exam.

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### 15+ Overlapping Schemes

Tamil Nadu runs 15+ welfare & scholarship schemes across ADW, BC/MBC Welfare, Pudhumai Penn, 7.5% Quota — each with hyper-specific thresholds across community, income, gender, and more.

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### Information Overload

A student cannot read 15 separate 40-page government gazette orders to know what they qualify for. There is no single unified source that aggregates all criteria.

⚠️

### The "Conflict Trap"

Students can't draw from two overlapping schemes simultaneously. Apply to both → **both get rejected**. Or they pick a ₹4K scheme when eligible for a ₹40K bundle.

THE SOLUTION

# SchemeMatch — An Algorithmic Scholarship Counselor

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### Zero-Barrier AI Profiling

Student types: *"I'm a girl from a TN govt school, doing B.Tech, 1.2 LPA, BC"* — our NL parser extracts all parameters instantly. Dual-mode: LLM primary + deterministic regex fallback.

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### 30+ Audited Scheme Catalog

19 Tamil Nadu + 11 National schemes, each coded with exact departmental criteria as deterministic `eligFn()` functions. Rule-based — zero hallucinations.

⚡

### Conflict-Aware Decision Engine — The Core Innovation

Doesn't dump 8 random schemes. Models schemes as a **weighted conflict graph**. Solves the **Maximum Weight Independent Set (MWIS)** problem to find the highest-value, legally valid, non-conflicting combination. Uses bitmask brute-force (n≤20, <50ms) with greedy fallback. AI only phrases results — **never decides eligibility**.

ALGORITHM PIPELINE

### From Natural Language to Optimal Aid Bundle in <50ms

Each scheme has `exclusiveGroup` and `conflictsWith` metadata mirroring real NSP "one benefit per fee line" rules.

NL Input → AI Parse → 30+ eligFn() → MWIS → ₹ Best Combo

SCALABILITY

# Built to Scale — From Tamil Nadu to All of India



### Declarative JSON Rule Schema

Scheme rules are **not hard-coded in logic**. Eligibility conditions, income ceilings, community gates, and conflict edges are parameterized in JSON. Adding a new scheme takes **2 minutes** — change a config file, no code redeployment needed.



### Auto-Ingestion Pipeline

Daily cron job crawls official government portals (HTML + PDF), extracts structured data via LLM, and proposes updates. **High-confidence extractions auto-apply**; medium/low go to human review queue with full audit trail.



### State-Agnostic Architecture

TN-specific categories (BC/MBC/DNC) are in the data layer, not the engine. Adding Karnataka, Maharashtra, or any state is just adding a new `stateSchemes.js` file with eligFn definitions — same engine, same conflict resolution.



### API-First Design

RESTful backend with `POST /schemes/match` and `POST /profile/parse` endpoints — can be plugged into TN e-District, UMANG, Naan Mudhalvan, or any govt portal with a single API webhook.

| 30+ | 2 min | 12 | 100% |
|---|---|---|---|
| Schemes Today | To Add a Scheme | REST API Endpoints | Offline Resilient |

IMPACT

# Real Impact — Measurable, Immediate, Equitable

**₹62,000**
Annual aid identified for a sample TN student profile

**Zero**
Rejections from double-dipping conflicts

**<50ms**
Full evaluation of 30+ schemes

**<1 sec**
Page load time on 3G connections

| FEATURE | SCHEMEMATCH | GOVT PORTAL | OTHER PROJECTS |
|---|---|---|---|
| Conflict Detection | ✅ MWIS Algorithm | ❌ None | ❌ None |
| Optimal Combination | ✅ Max-value subset | ❌ Flat list only | ❌ Flat list only |
| Natural Language Input | ✅ AI + regex fallback | ❌ Complex forms | ⚠️ AI only (no fallback) |
| Offline Capability | ✅ 100% client-side | ❌ Server-dependent | ❌ API-dependent |
| Auto Data Freshness | ✅ Ingestion pipeline | ❌ Manual updates | ❌ Static data |

REFERENCES

# Data Sources & Technology References

### Adi Dravidar & Tribal Welfare Dept, TN
tnadw.tn.gov.in
SC/ST scholarship schemes, eligibility criteria

### BC, MBC & Minorities Welfare Dept, TN
bcw.tn.gov.in
BC/MBC/DNC post-matric & welfare schemes

### National Scholarship Portal (NSP)
scholarships.gov.in
Central govt schemes, "one benefit per fee line" rule

### AICTE Official Portal
aicte-india.org
Pragati & Saksham scholarship criteria

### TN Govt Scheme Catalog
tn.gov.in/scheme_beneficiary_list.php
488+ statewide schemes (filtered to education)

### DST INSPIRE & CSIR-UGC NET
indiascienceandtechnology.gov.in
Science & research fellowship criteria

## TECHNOLOGY STACK

HTML5 / CSS3 / Vanilla ES6+ | Node.js + Express | MongoDB Atlas + Mongoose | JWT + bcrypt Auth | OpenRouter LLM API | MWIS (Bitmask Brute Force) | Vercel Deployment | GitHub Actions CI/CD | node-cron Ingestion

## Thank You — We're Ready for Your Questions!

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